Anatomy of a Wireless Earbud

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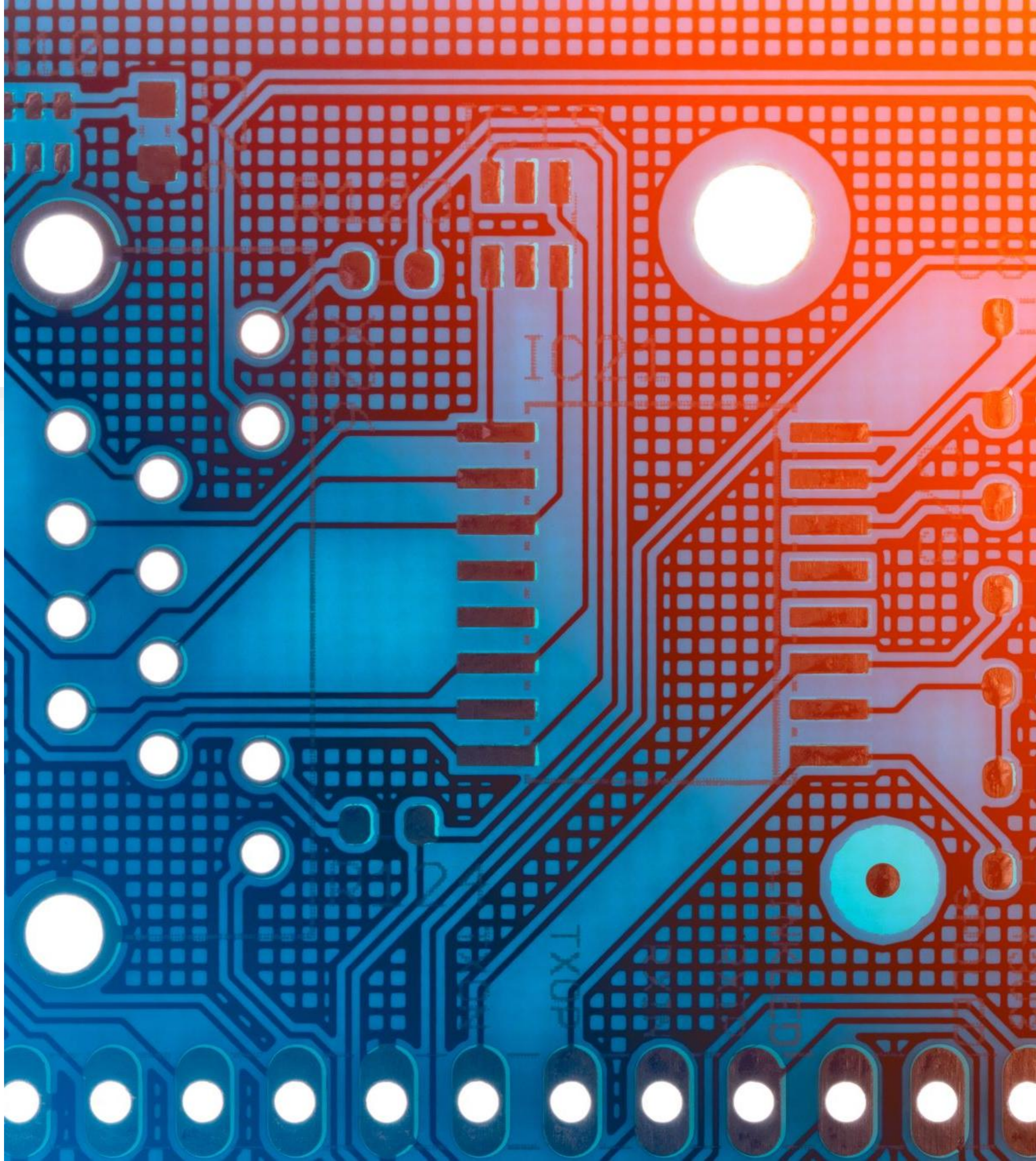
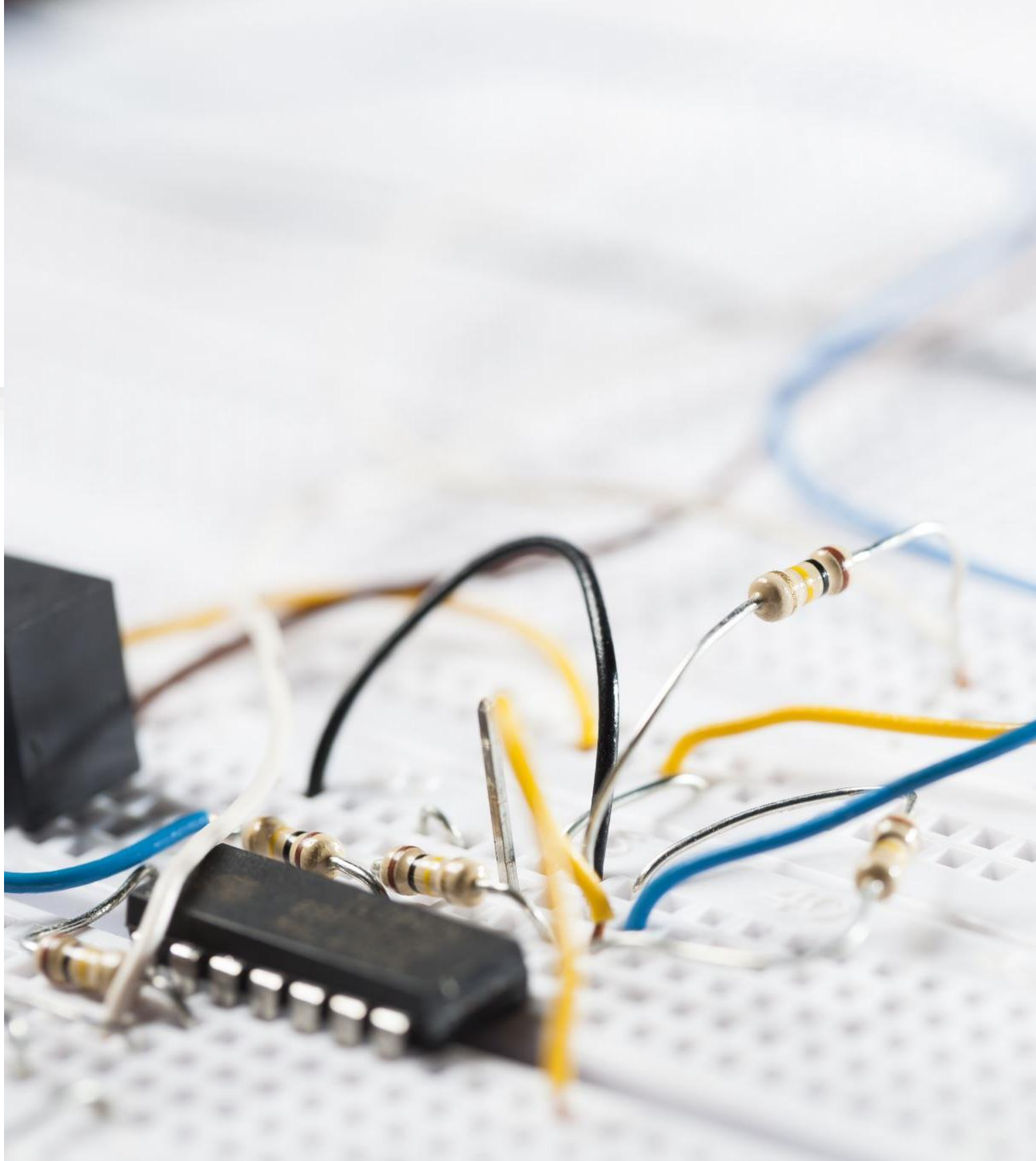
Introduction to Wireless Earbuds:

- Wireless earbuds (TWS – True Wireless Stereo) are compact Bluetooth audio devices.
- They contain multiple electronic systems in a very small space.
- Purpose of teardown:
  - Understand internal hardware
  - Study circuit design
  - Identify electronic components
- This project focuses on the **main PCB (Printed Circuit Board)** of an earbud.



Aim of This Study:

- To dismantle a non-working earbud safely
- To analyze internal PCB and components
- To understand:
  - Power system
  - Charging circuit
  - Audio system
  - Control system

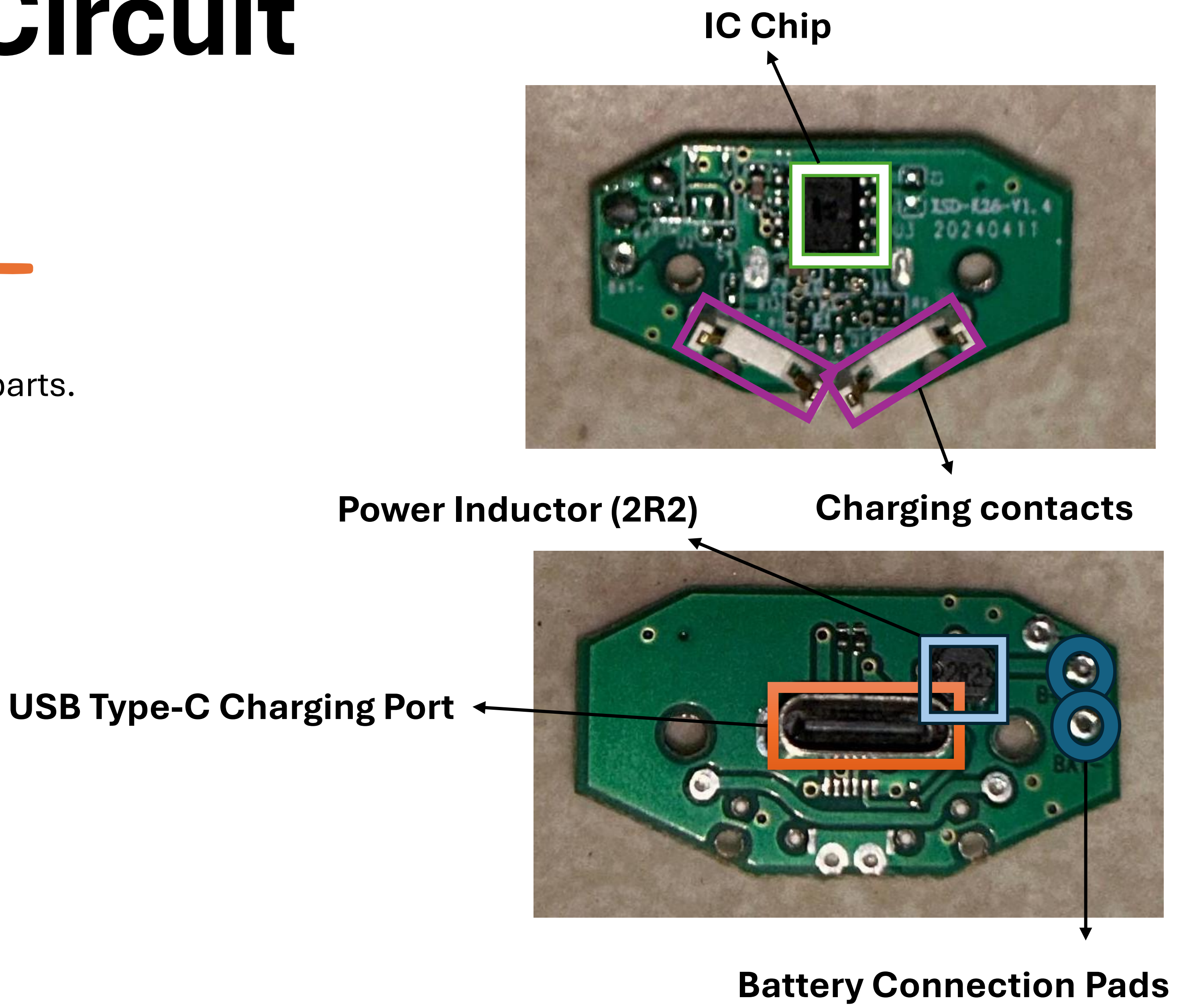


Overview of Components Found:

- PCB (Circuit Board)
- Bluetooth SoC Chip
- Charging Circuit
- Voltage Regulator
- Lithium-ion Battery
- Connectors & Contacts
- Passive Components (Resistors, Capacitors)
- Metal Springs & Contacts
- Screws
- Plastic Housing

PCB (Printed Circuit Board):

- **What it is:**
  Green board that connects all electronic parts.
- **Function:**
- Carries electrical signals
- Supplies power between components
- **Real-life example:**
  Motherboard in a laptop
- **Applications:**
- Smartphones
- Smart watches



Bluetooth SoC (Main IC):

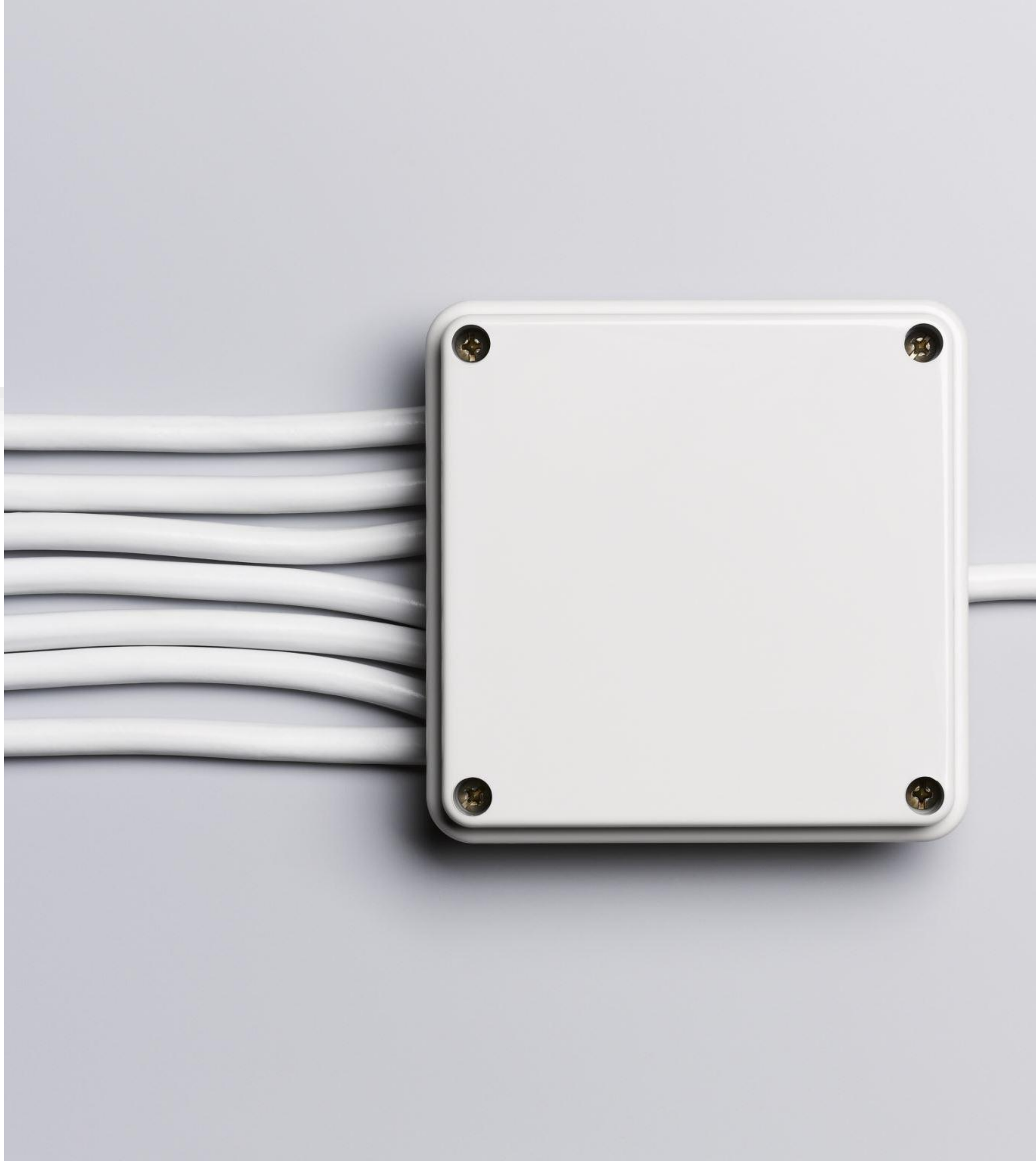
- **What it is:** The biggest black chip on PCB — the *brain*.
- **Function:**
- Bluetooth communication
- Audio processing
- Controls mic & speaker
- **Real-life example:** Processor in a smartphone
- **Applications:**
- Wireless headphones
- Bluetooth speakers

Charging IC:

- **What it is:** Small chip near charging port.
- **Function:**
- Charges battery safely
- Prevents overcharging
- **Real-life example:** Phone battery charging controller
- **Applications:**
- Power banks
- Smart watches

Voltage Regulator:

- **What it is:** Tiny IC regulating voltage.
- **Function:**
- Converts 3.7V battery → safe voltages
- Protects sensitive electronics
- **Real-life example:** Voltage stabilizer in TV
- **Applications:**
- Routers
- Arduino boards



Lithium-ion Polymer Battery:

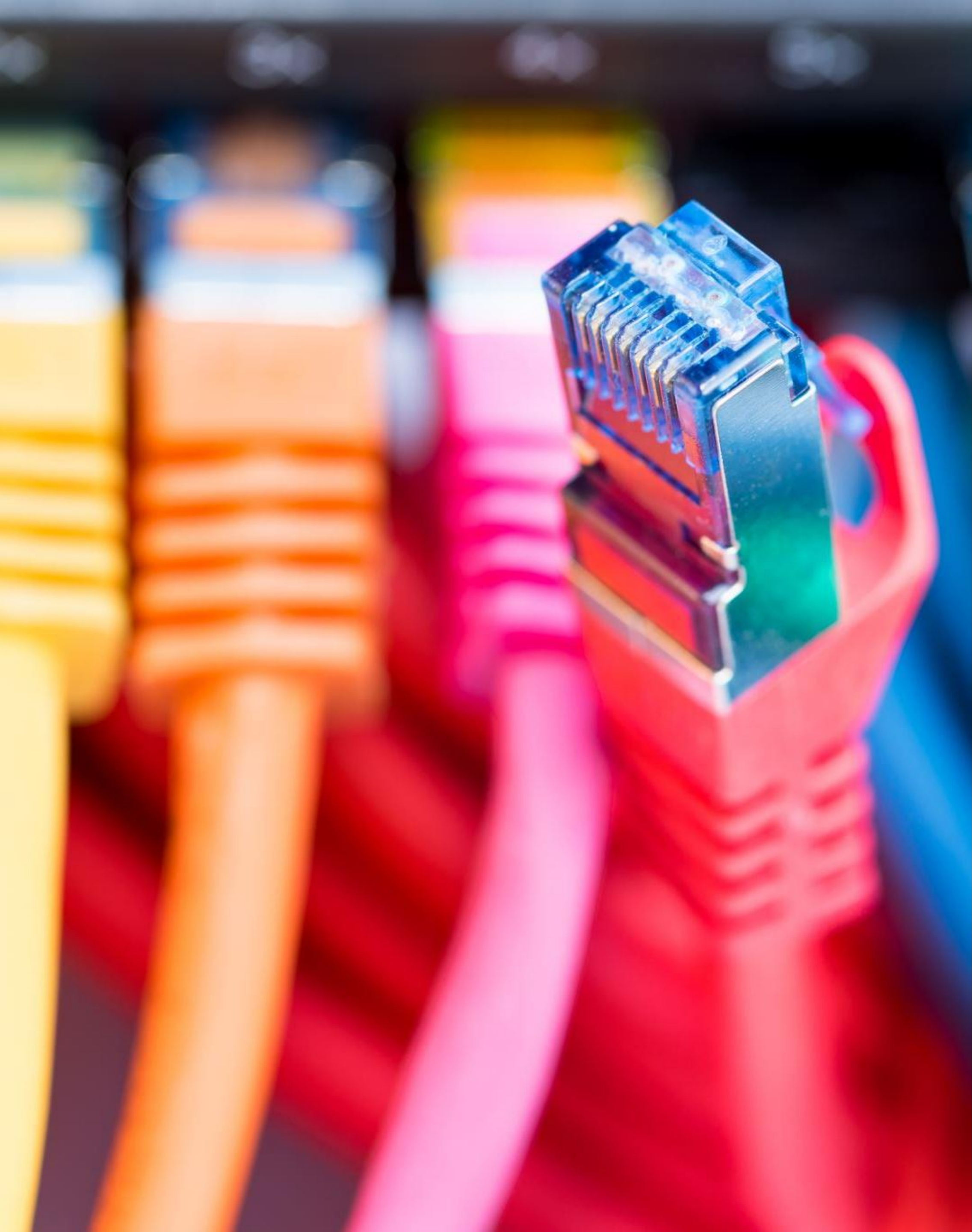
- **What it is:**
  Small flat rechargeable battery.
- **Function:**
  Stores power for earbud
- **Real-life example:**
  Phone battery
- **Applications:**
- Drones
- Fitness bands

Resistors:

- **Function:**
- Control current
- Set voltage levels
- **Real-life example:**
  Water tap controlling water flow
- **Applications:**
- LED circuits
- Power supplies

Capacitors:

- **Function:**
- Store charge temporarily
- Remove noise from power
- **Real-life example:** Water tank storing water
- **Applications:**
- Audio amplifiers
- TV power boards



Connectors:

- **What they do:**
  Connect speaker, battery, mic.
- **Real-life example:**
  USB cable plug
- **Applications:**
- Laptop internal cables
- Mobile screen connectors

Charging Contacts / Metal Springs:

- **Function:**
- Touch charging case terminals
- Transfer electricity
- **Real-life example:**
  Battery contacts in remote
- **Applications:**
- Charging docks
- Battery compartments

Screws:

- **Function:**
  Hold internal parts in position
- **Real-life example:**
  Screws in spectacles
- **Applications:**
- Laptops
- Home appliances

Plastic Housing:

Function:

- Protects electronics
- Ergonomic design
- **Real-life example:** Phone back cover
- **Applications:**
- Headphones
- Remote controls

Working Principle:

- Power & Charging Flow : Charging Case ➝ Metal Contacts ➝ Charging IC ➝ Li-ion Battery ➝ Voltage Regulator ➝ Bluetooth SoC
- Music Playback Flow : Smartphone ➝ *(Bluetooth Signal)* ➝ Bluetooth SoC ➝ Audio Amplifier ➝ Speaker Driver ➝ Sound to Ear
- Voice Call Flow : Your Voice ➝ Microphone ➝ Bluetooth SoC ➝ *(Bluetooth Transmission)* ➝ Smartphone
- Control Flow : Touch/Button Input ➝ SoC Processing ➝ Command Action *(Play/Pause/Call/Volume)*

Step-by-Step Connection Process:

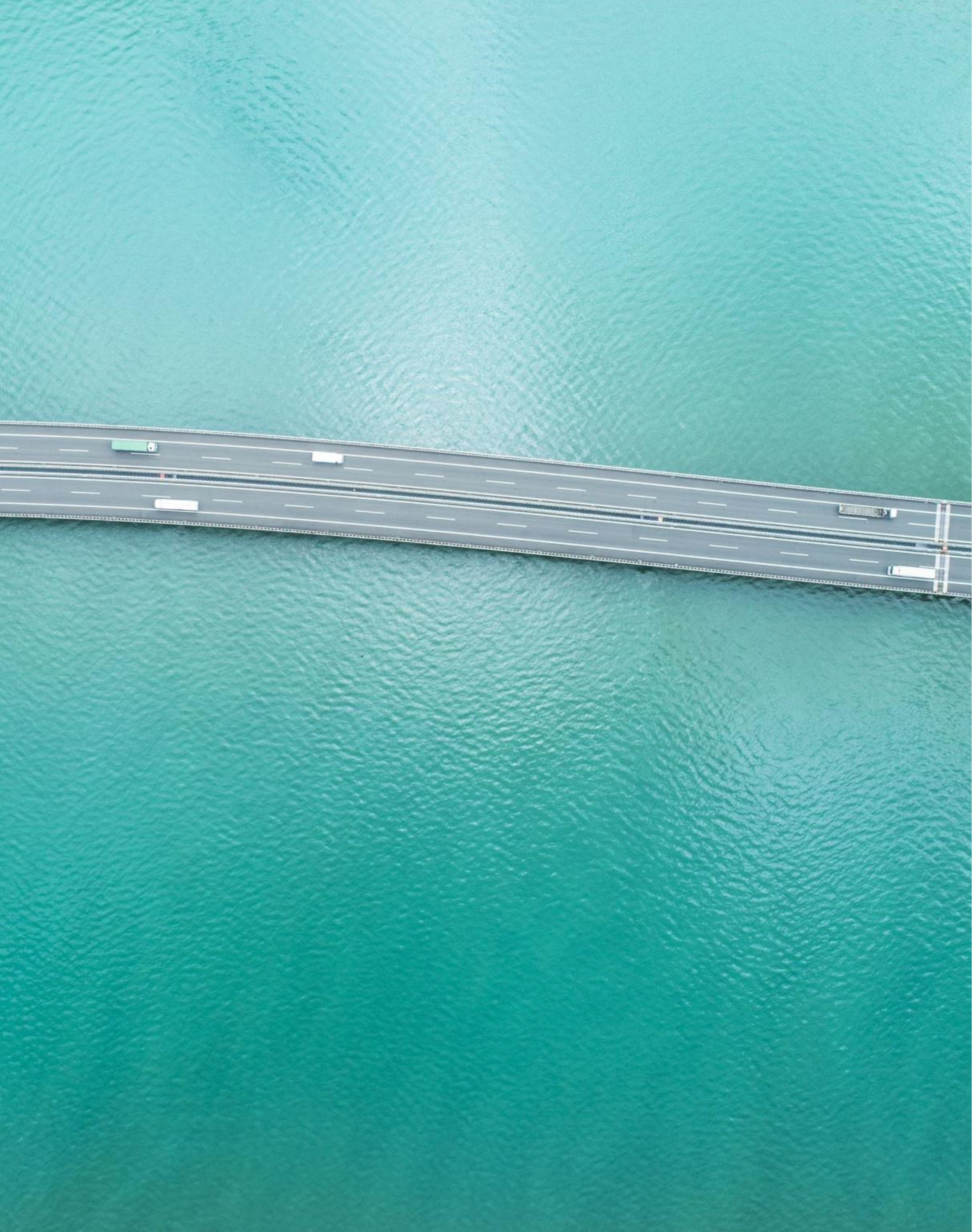
- 1️⃣ **Earbuds Turn On :** When you open the case, power reaches the Bluetooth chip inside the earbuds.
- 2️⃣ **Earbuds Send Signals :** The chip sends out small radio signals through a tiny antenna, saying: *"I’m here and ready to connect."*
- 3️⃣ **Phone Detects the Signal :** Your phone’s Bluetooth is always listening. It receives this signal and recognizes the earbuds.
- 4️⃣ **Pairing Happens :** Phone and earbuds exchange identification data and create a secure connection.
- 5️⃣ **Audio Transfer Starts :** Phone converts music into digital data, Data is sent as radio waves, Earbuds receive it, Earbuds convert it back into sound.

What Are These Signals?

- They are **electromagnetic radio waves**:
- Travel through air
- Invisible to our eyes
- Work up to ~10 meters

Conclusion:

- The **PCB** connects and supports all electronic components.
- The **Bluetooth SoC** acts as the brain, handling pairing, signal processing, and device control.
- The **Lithium-ion battery** powers the system, while the **charging IC** ensures safe charging.
- The **voltage regulator** provides stable voltage to protect sensitive circuits.
- The **speaker and microphone** enable two-way sound communication.
- **Resistors and capacitors** maintain current control and noise filtering.
- **Final Understanding:** A wireless earbud is not just a sound device — it is a **mini smart system** combining electronics, signal processing, and wireless technology in a highly efficient and space-saving design.



THANK YOU